Week 5: Mini-Project — Simple Quiz Game

Goal

Build a quiz game that asks questions, checks answers, tracks the score, and gives a final grade. This project combines everything from this week: loops, conditionals, comparison operators, and the accumulator pattern.

This is what separates someone who reads about code from someone who **builds things**.

Setup

Start with this starter code:

```
// Quiz questions — each has a question, options, and the correct answer
let questions = [
{
question: "What is the capital of Nigeria?",
options: "A) Lagos  B) Abuja  C) Kano  D) Bauchi",
answer: "B"
},
{
question: "What does HTML stand for?",
options: "A) How To Make Links  B) HyperText Markup Language  C) Home Tool Markup
Language  D) Hyper Transfer Main Language",
answer: "B"
},
{
question: "Which operator means strict equality in JavaScript?",
options: "A) ==  B) =  C) ===  D) !=",
answer: "C"
},
{
question: "What is 2 + 2 * 3 in JavaScript?",
options: "A) 12  B) 8  C) 10  D) 6",
answer: "B"
},
{
question: "Which of these is a falsy value?",
options: "A) []  B) 'false'  C) 0  D) ' '",
answer: "C"
}
];
```

```
// Simulated player answers (as if someone typed these)
let playerAnswers = ["B", "B", "A", "B", "C"];
```

The 6 Challenges

For each challenge, write 4 sections:

1. **Understand** — What are the inputs? What should the output be? Any edge cases?
2. **Plan** — Write pseudocode (no JavaScript yet)
3. **Execute** — Write the JavaScript code
4. **Reflect** — What could go wrong? How could you improve it?

Challenge 1: Loop Through Questions

Use a `for` loop to print each question number and the question text.

Expected output:

```
Question 1: What is the capital of Nigeria?
Question 2: What does HTML stand for?
...
```

Solution with UPER

Understand:
- Input: Array of question objects
- Output: Each question printed with its number
- Edge case: What if the array is empty?

Plan:
```
FOR i FROM 0 TO length of questions - 1
PRINT "Question " + (i + 1) + ": " + questions[i].question
ENDFOR
```

Execute:
```javascript
for (let i = 0; i  questions.length; i++) {
console.log("Question " + (i + 1) + ": " + questions[i].question);
console.log("  " + questions[i].options);
console.log("");
}
```

Reflect: We use `(i + 1)` because arrays start at index 0, but humans count from 1. The empty `console.log("")` adds spacing between questions for readability.

Challenge 2: Check Answers Using Strict Equality

Compare each player answer to the correct answer using `===`. Print whether each answer is correct or wrong.

Expected output:

```
Question 1: Correct!
Question 2: Correct!
Question 3: Wrong! The answer was C
Question 4: Correct!
Question 5: Correct!
```

Solution with UPER

Understand:
- Input: `questions` array (with correct answers) and `playerAnswers` array
- Output: "Correct!" or "Wrong! The answer was X" for each question
- Edge case: What if arrays have different lengths?

Plan:
```
FOR i FROM 0 TO length of questions - 1
IF playerAnswers[i] EQUALS questions[i].answer THEN
PRINT "Question " + (i+1) + ": Correct!"
ELSE
PRINT "Question " + (i+1) + ": Wrong! The answer was " + questions[i].answer
ENDIF
ENDFOR
```

Execute:
```javascript
for (let i = 0; i  questions.length; i++) {
if (playerAnswers[i] === questions[i].answer) {
console.log("Question " + (i + 1) + ": Correct!");
} else {
console.log("Question " + (i + 1) + ": Wrong! The answer was " + questions[i].answer);
}
}
```

Reflect: We use `===` (strict equality) to compare strings. Using `==` would also work here since both sides are strings, but `===` is the professional standard and safer habit.

Challenge 3: Keep Score with an Accumulator

Add a score counter that starts at 0 and increases by 1 for each correct answer. Print the final score.

Expected output:

```
You scored 4 out of 5
```

Solution with UPER

Understand:
- Input: Both arrays (same as Challenge 2)
- Output: Total score as "X out of Y"
- Edge case: Score could be 0 (all wrong) or equal to total (all correct)

Plan:
```
SET score TO 0
FOR i FROM 0 TO length of questions - 1
IF playerAnswers[i] EQUALS questions[i].answer THEN
ADD 1 TO score
ENDIF
ENDFOR
PRINT "You scored " + score + " out of " + length of questions
```

Execute:
```javascript
let score = 0;

for (let i = 0; i  questions.length; i++) {
if (playerAnswers[i] === questions[i].answer) {
score++;
}
}

console.log("You scored " + score + " out of " + questions.length);
// Output: "You scored 4 out of 5"
```

Reflect: This is the accumulator pattern — start at 0, add 1 each time the condition is true. `score++` is shorthand for `score = score + 1`.

Challenge 4: Calculate the Grade

Convert the score to a percentage and assign a letter grade using `if/else if/else`.

- 80-100%: “A“
- 60-79%: “B“
- 40-59%: “C“
- Below 40%: “F“

Expected output:

```
Score: 80% — Grade: A
```

Solution with UPER

Understand:
- Input: Score (number) and total questions (number)
- Output: Percentage and letter grade
- Edge case: Division by zero if no questions (but we have 5)

Plan:
```
SET percentage TO (score / total) * 100
IF percentage >= 80 THEN grade = "A"
ELSE IF percentage >= 60 THEN grade = "B"
ELSE IF percentage >= 40 THEN grade = "C"
ELSE grade = "F"
PRINT percentage + "% — Grade: " + grade
```

Execute:
```javascript
let score = 4;
let total = 5;
let percentage = (score / total) * 100;
let grade;

if (percentage >= 80) {
grade = "A";
} else if (percentage >= 60) {
grade = "B";
} else if (percentage >= 40) {
grade = "C";
} else {
grade = "F";
}

console.log("Score: " + percentage + "% — Grade: " + grade);
// Output: "Score: 80% — Grade: A"
```

Reflect: 4 out of 5 = 80%, which is right at the boundary for an "A". Change `score` to 3 to see "60% — Grade: B". Boundary values are important to test.

Challenge 5: Put It All Together

Combine challenges 1-4 into one complete quiz game. The program should:

1. Print each question with its options
2. Check each answer and say correct/wrong
3. Keep a running score
4. Print the final score and grade at the end

Solution with UPER

Understand:
- Input: questions array, playerAnswers array
- Output: Full quiz experience — questions, feedback, score, grade
- Edge case: Player might answer all correctly or all wrong

Plan:
```
SET score TO 0
FOR i FROM 0 TO length of questions - 1
PRINT question number and text
PRINT options
PRINT player's answer
IF answer is correct THEN
PRINT "Correct!"
ADD 1 TO score
ELSE
PRINT "Wrong! Answer was " + correct answer
ENDIF
PRINT blank line
ENDFOR
SET percentage TO (score / total) * 100
DETERMINE grade from percentage
PRINT final score and grade
```

Execute:
```javascript
let score = 0;

console.log("=== QUIZ GAME ===");
console.log("");

for (let i = 0; i  questions.length; i++) {
console.log("Question " + (i + 1) + ": " + questions[i].question);
console.log("  " + questions[i].options);
console.log("  Your answer: " + playerAnswers[i]);

if (playerAnswers[i] === questions[i].answer) {
console.log("  Correct!");
score++;
```

```
} else {
console.log("  Wrong! The answer was " + questions[i].answer);
}
console.log("");
}

// Calculate grade
let percentage = (score / questions.length) * 100;
let grade;

if (percentage >= 80) {
grade = "A";
} else if (percentage >= 60) {
grade = "B";
} else if (percentage >= 40) {
grade = "C";
} else {
grade = "F";
}

console.log("=== RESULTS ===");
console.log("Score: " + score + "/" + questions.length + " (" + percentage + "%)");
console.log("Grade: " + grade);

if (grade === "A") {
console.log("Excellent work!");
} else if (grade === "B") {
console.log("Good job! Keep studying.");
} else if (grade === "C") {
console.log("You passed, but review the material.");
} else {
console.log("Keep practicing. You'll get there!");
}
```

Reflect: This is a real, complete program. It loops, branches, accumulates, and produces meaningful output. You could extend it by adding more questions, randomizing the order, or adding a timer. Every concept from this week is used here.

Challenge 6 (Bonus): Add a Pass/Fail Summary with Switch

Use a `switch` statement to print a message based on the grade. Also use a `do...while` loop to let the player “retake“ the quiz (simulate with a second set of answers).

Solution with UPER

Understand:

- Input: Grade letter from Challenge 5
- Output: Custom message per grade using switch, plus a retake loop
- Edge case: What if grade doesn't match any case? Use default.

Plan:
```
SWITCH grade
CASE "A": PRINT celebration message
CASE "B": PRINT encouragement
CASE "C": PRINT study tip
CASE "F": PRINT retry message
DEFAULT: PRINT error

Simulate retake with do...while
```

Execute:
```javascript
// Grade message using switch
switch (grade) {
case "A":
console.log("You're a star! Share your score in the WhatsApp group.");
break;
case "B":
console.log("Solid performance! Review the ones you missed.");
break;
case "C":
console.log("You passed! Re-read the lesson and try again.");
break;
case "F":
console.log("Don't give up! Review the lesson and retake.");
break;
default:
console.log("Something went wrong with grading.");
}

// Simulate retake with do...while
let allAttempts = [
["B", "B", "A", "B", "C"],  // First attempt: 4/5
["B", "B", "C", "B", "C"]   // Retake: 5/5
];
let attemptIndex = 0;

do {
let currentAnswers = allAttempts[attemptIndex];
let attemptScore = 0;

for (let i = 0; i  questions.length; i++) {
if (currentAnswers[i] === questions[i].answer) {
attemptScore++;
}
```

}

let attemptPercent = (attemptScore / questions.length) * 100;
console.log("Attempt " + (attemptIndex + 1) + ": " + attemptScore + "/" + questions.length + " (" + attemptPercent + "%)");

attemptIndex++;
} while (attemptIndex allAttempts.length);
```

**Reflect:** The `do...while` guarantees the quiz runs at least once. The `switch` is cleaner than an `if/else if` chain when matching specific string values. Both are real patterns you'll use in bigger programs.

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## How to Submit

1. Write all 6 challenges in a single file (bonus challenge is optional)
2. Include your UPER documentation as comments above each solution
3. Take a screenshot showing the full output in your console
4. Upload to Google Classroom

**Example format:**

```
// === Challenge 1: Loop Through Questions ===
// Understand: Input is array of question objects, output is numbered list
// Plan: Loop from 0 to length-1, print index+1 and question text
// (your code here)
// Reflect: Used i+1 for human-friendly numbering

// === Challenge 2: Check Answers ===
// ...
```
```